Math 344, F18 **Name:** Answer

Quiz 9

<u>Instructions:</u> **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (10 points) Find the best fitting line $y = C + Dt$ for $b = 1, 2, 4$ at times $t = -1, 0, 2$.

(a) Write down the resulting equation $\mathbf{Ax} = \mathbf{b}$.

$$\underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} C \\ D \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}}_{b}$$

(b) Solve the normal euqation for the least squares fitting, $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$, to find C and D.

$$A^T A \hat{x} = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

Solve

$$\begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{cases} C = 2 \\ D = 1 \end{cases}$$

Thus the least squares fitting line is

$$y = 2 + t$$